

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

```
model = GaussianNB()
```

```
model.fit(X_train, y_train)
```

▼ GaussianNB ⓘ ?

GaussianNB()

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Actual values:  ", y_test)
print("Predicted values:", y_pred)
print(f"\nAccuracy: {accuracy * 100:.2f}%")
```

```
Actual values:  [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Predicted values: [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 2 1 0 2 0]
```

```
Accuracy: 96.67%
```

```
display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
)

display.plot()
plt.title("Naïve Bayes Confusion Matrix")
plt.show()
```

